ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Section A: objective questions (30) · Section B: hydraulic calculations, pneumatic sequencing, PID control and system response, servo systems, condition monitoring and reliability, and the design of a pick-and-place unit (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. Use π = 3,142 where needed.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Pressure applied to a confined fluid is transmitted equally in all directions. This is …

A Bernoulli's principle.
B Pascal's law.
C Boyle's law.
D Hooke's law.

1.1.2 The force produced by a hydraulic cylinder is calculated with …

A F = p ÷ A
B F = A ÷ p
C F = p × A
D F = m ÷ a

1.1.3 In the sequence A+ B+ B− A−, "A+" means …

A cylinder A extends.
B cylinder A retracts.
C valve A closes.
D pump A starts.

1.1.4 In a PID controller, the term that removes steady-state error is the …

A proportional term.
B integral term.
C derivative term.
D feedback sensor.

1.1.5 The amount by which a system's output goes past the set point is called …

A settling time.
B rise time.
C dead band.
D overshoot.

1.1.6 A servo system is a …

A closed-loop position or speed control system.
B open-loop system.
C hydraulic pump.
D type of sensor only.

1.1.7 Measuring vibration to detect developing faults in a machine is an example of …

A breakdown maintenance.
B corrective maintenance only.
C condition monitoring.
D commissioning.

1.1.8 MTBF stands for …

A maximum torque before failure.
B motor test before fitting.
C mean time between failures.
D minimum time between fixes.

1.1.9 A thermographic (infrared) camera is used in maintenance to find …

A hot spots such as loose connections or bad bearings.
B air leaks only.
C program errors.
D the PLC address.

1.1.10 A virtual copy of a real machine that is updated with live sensor data is a …

A prototype only.
B ladder diagram.
C block diagram.
D digital twin.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Hydraulic systems can produce much larger forces than pneumatic systems of the same size.

1.2.2 A cylinder retracts with less force than it extends because of the piston rod area.

1.2.3 Oil is compressible, which is why hydraulic systems are springy.

1.2.4 A cascade circuit is used to avoid signal conflict in pneumatic sequences.

1.2.5 Increasing the proportional gain too much can make a system oscillate.

1.2.6 The derivative term reacts to how fast the error is changing.

1.2.7 Steady-state error is the difference between the set point and the output after the system has settled.

1.2.8 Predictive maintenance replaces parts only after they have failed.

1.2.9 Availability increases when the mean time to repair (MTTR) decreases.

1.2.10 An encoder is often used as the feedback device in a servo motor.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Valve that limits the maximum pressure in a hydraulic system	A	pressure relief valve
1.3.2	Valve that controls the speed of a cylinder	B	preventive maintenance
1.3.3	Diagram that shows the movement of each cylinder step by step	C	MTTR
1.3.4	Desired value that the controller tries to reach	D	motion profile
1.3.5	Set point minus measured value	E	error
1.3.6	Time for the output to stay within a small band around the set point	F	commissioning
1.3.7	Average time taken to repair a machine	G	set point
1.3.8	Maintenance done at fixed intervals to prevent breakdowns	H	displacement-step diagram
1.3.9	Testing a new system step by step before handing it over	I	settling time
1.3.10	Graph of speed against time for a planned movement	J	flow control valve

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: HYDRAULICS AND PNEUMATICS

2.1 A hydraulic cylinder has a piston diameter of 80 mm and a rod diameter of 40 mm. The system pressure is 10 MPa.

2.1.1 Calculate the extending force. (4)

2.1.2 Calculate the retracting force. (4)

2.1.3 The pump delivers 20 litres per minute. Calculate the extending speed of the piston in m/s. (4)

2.2 Two pneumatic cylinders must operate in the sequence A+ B+ A− B−. Draw the displacement-step diagram. (4)

2.3 Give TWO advantages of hydraulics and TWO advantages of pneumatics for robot actuators. (4)

[20]

QUESTION 3: CONTROL AND SERVO SYSTEMS

3.1 Explain the effect of EACH term of a PID controller (P, I and D) on the response of a motor-speed system. (6)

3.2 Sketch the step response of an under-damped system and label the set point, overshoot, rise time, settling time and steady-state error. (5)

3.3 A servo axis has a set point of 200 mm and settles at 196 mm. Calculate the steady-state error and the percentage error. (3)

3.4 Explain why a trapezoidal motion profile (accelerate, constant speed, decelerate) is used for robot axes instead of starting and stopping suddenly. (3)

[17]

QUESTION 4: CONDITION MONITORING AND RELIABILITY

4.1 Differentiate between breakdown, preventive and predictive (condition-based) maintenance. (6)

4.2 Name THREE condition-monitoring techniques and the fault each one detects. (6)

4.3 A robot cell ran for 2 000 hours and had 4 failures. The total repair time was 20 hours.

4.3.1 Calculate the MTBF. (2)

4.3.2 Calculate the MTTR. (2)

4.3.3 Calculate the availability as a percentage using A = MTBF ÷ (MTBF + MTTR) × 100. (2)

[18]

QUESTION 5: DESIGN OF A PICK-AND-PLACE UNIT

5.1 A pick-and-place unit must take a part from a feeder and place it in a fixture using a horizontal and a vertical axis and a gripper.

5.1.1 Recommend an actuator for EACH axis and for the gripper, with reasons. (6)

5.1.2 Name FOUR sensors needed and the purpose of each. (4)

5.1.3 Name TWO documents that must be handed over with the finished system, and say why each is needed. (2)

5.1.4 Explain how a digital twin could help during the design and commissioning of this unit. (3)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

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END OF QUESTION PAPER